

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

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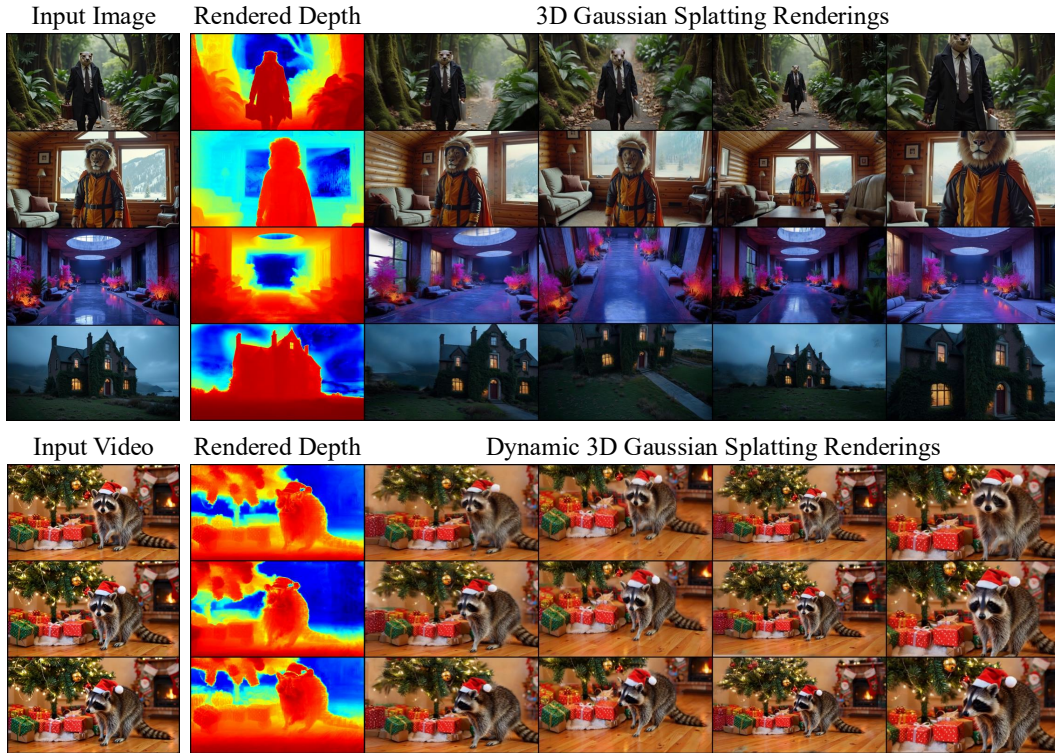


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** From a single image (top), and across a wide variety of scenes, *Lyra* infers a 3D Gaussian Splatting (3DGS) representation in a feed-forward fashion by self-distilling a video diffusion model, without requiring any real-world multi-view data. Given a video input (bottom), the same model infers a dynamic 3DGS that offers interactive control in both time (rows) and viewpoint (columns).

ABSTRACT

The ability to generate virtual environments underpins a broad spectrum of applications, spanning entertainment and gaming as well as physical AI domains such as robotics, autonomous driving, and industrial AI, and it naturally extends to any setting that benefits from controllable synthetic 3D content. Current learning-based 3D reconstruction methods generally depend on captured real-world multi-view data, a requirement that is rarely satisfied at the scale and diversity that broad deployment demands. Recent advancements in video diffusion models have shown remarkable imagination capabilities across a wide range of scenes, yet their inherently 2D nature limits their use in the many downstream settings—such as simulation, where an agent must navigate and interact with its surroundings—that require an explicit spatial representation. In this paper, we propose a general self-distillation framework that distills the implicit 3D knowledge embedded in video diffusion models into an explicit 3D Gaussian Splatting (3DGS) representation, removing the dependence on multi-view training data. Specifically, we augment the typical RGB decoder with a 3DGS decoder that is supervised directly by the

output of the RGB decoder. Under this design, the 3DGS decoder can be trained entirely from synthetic data generated by video diffusion models, without tying the approach to any particular capture setup or scene type. At inference time, our model synthesizes 3D scenes from either a text prompt or a single image and renders them in real time. The same framework further generalizes to dynamic 3D scene generation from a monocular input video. Across static and dynamic settings, our experiments show that the framework attains state-of-the-art 3D scene generation quality, and its design suggests applicability well beyond the specific scenarios we evaluate.

1 INTRODUCTION

Creating high-quality 3D environments at scale is a long-standing challenge in computer vision and graphics, and it underlies a wide array of applications—from film-making, VR/AR, and physical AI with closed-loop simulation to any workflow that consumes interactive 3D content. What these applications share is a need for explicit 3D representations that support real-time rendering, physical interaction, and consistent multi-view synthesis.

Recent advances in neural 3D reconstruction (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) enable recovering such representations from posed images. However, their reliance on accurate camera poses and high-quality images significantly limits how broadly they can be scaled. The challenge is even greater for dynamic scenes, which typically require synchronized multi-camera setups (Li et al., 2022). Reconstruction methods are also inherently constrained to the observed content and cannot extrapolate beyond the input views. Recent works aim to mitigate these limitations with feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c), but these approaches face a bottleneck of their own: the scarcity of diverse, large-scale 3D training data, which leads to poor generalization once scenes fall outside the training distribution.

Video diffusion models (Cosmos, 2025; Wan, 2025) offer a promising and broadly applicable alternative. Trained on massive internet-scale video corpora, they achieve impressive fidelity and generalize across an extremely wide range of environments. By learning from real-world videos with varied camera trajectories—ranging from handheld motion to cinematic panning and drone footage—these models implicitly encode cues about the underlying 3D world without requiring any multi-view training data. Unlike reconstruction approaches (Charatan et al., 2024), generative models can also hallucinate plausible content beyond what is visible in the input frames. Yet, video diffusion models produce only 2D frames and lack an explicit 3D representation. This limits their use in simulation and the many downstream tasks that demand geometric consistency, long-term coherence, and physical interaction. Encouragingly, recent work (Wang et al., 2024e; Ren et al., 2025) has shown that video models can be adapted for explicit camera control using only relatively small-scale posed datasets. This ability to generate posed image sequences turns them into a versatile tool that can serve as both input and supervision for 3D reconstruction models across diverse scenarios.

Motivated by this insight, we bridge these two paradigms, reconstruction and generation, through what we call *generative 3D scene reconstruction*. We introduce *Lyra*, a general method for generating explicit 3D environments from the latent representations of video diffusion models in a single forward pass. Crucially, *Lyra* is trained in a self-distillation framework (Fig. 2), with a 3D Gaussian Splatting (3DGS) decoder operating directly in the latent space of a video diffusion model. Given a single input image or video, we sample a camera trajectory as conditioning input, denoise the resulting video latent, and decode it along two parallel branches. The first branch uses the standard RGB decoder to synthesize a video sequence, while the second is our 3DGS decoder, which produces an explicit 3D representation. Together, these branches form a self-distillation framework in which the RGB branch (teacher) supervises the 3DGS branch (student). To broaden viewpoint coverage, we sample multiple camera trajectories, generate multiple video latents, and train the 3DGS decoder to fuse information across them while mitigating long-term and multi-view inconsistencies.

This generative reconstruction framework offers three key benefits that hold across a wide range of scene types: **(i)** it enables generation of large-scale synthetic environments spanning highly diverse scenarios directly from video diffusion models, removing the need for real-world multi-view captures; **(ii)** by operating in the latent space of a video model, it processes many views efficiently, without the heavy memory overhead of pixel-space feed-forward reconstruction methods; and **(iii)** its explicit 3DGS output guarantees geometric consistency, yielding representations that transfer directly to

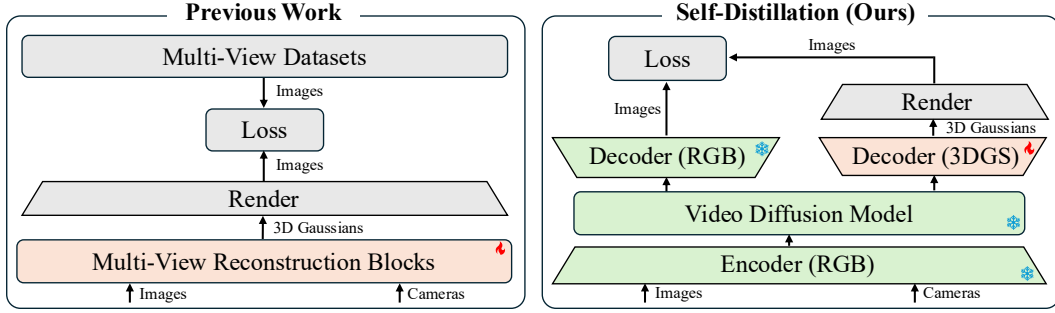


Figure 2: **Self-distillation framework of Lyra.** Previous work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks on real-world datasets of limited diversity (Zhou et al., 2018; Ling et al., 2024b). In contrast, we propose a general self-distillation framework (right) for generative 3D scene reconstruction: a pre-trained camera-controlled video diffusion model, through its RGB decoder output (teacher), supervises the rendering of the 3DGS decoder (student). Because the teacher is pre-trained on broad, diverse 2D video data, it can supply multi-view supervision spanning a wide range of scenes.

downstream tasks such as physical simulation and, more broadly, any application that requires interactive 3D content. Consequently, at inference time, Lyra generates high-quality 3D Gaussian scenes from monocular input that support real-time rendering, *without requiring any additional optimization or post-processing*.

We further extend the same self-distillation framework to dynamic 4D generation from monocular video, illustrating that the approach is not specific to static content. In this setting, the video model (teacher) provides space–time supervision, while the student learns to produce time-conditioned 3DGS representations that enable novel-view synthesis of dynamic scenes.

Overall, our work makes the following contributions:

- We introduce a general self-distillation framework that trains a 3DGS student decoder using a pre-trained camera-controlled video diffusion model as the teacher, eliminating the need for captured multi-view real-world data.
- The same framework extends naturally to dynamic scenes, supporting 4D reconstruction from a monocular video.
- Our model generalizes across a broad diversity of scenes, achieving state-of-the-art results in both single-image 3D scene generation and single-video 4D scene generation.

2 RELATED WORK

Multi-view image generation. Early works on multi-view image generation (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) mainly focus on object-centric scenes without background. Follow-up works (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) broaden multi-view image generators to the full scene scale. CAT3D (Gao et al., 2024b) extends a pre-trained image diffusion model to multi-view image generation from any generated or real image input. In a subsequent stage, the generated multi-view images are reconstructed into an explicit 3D representation, e.g., NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023). These methods have further been extended to generate dynamic 3D scenes from multiple viewpoints using space-time diffusion models (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). Even though these methods demonstrate impressive results, grounding the generations in an explicit 3D representation requires an expensive optimization stage that is not amortized across scenes. Rather than creating multi-view images and then reconstructing them by optimization, we aim to *directly* generate a 3D scene from text or a single image, in a manner that applies across scene types.

Camera-conditioned video generation. There has been significant progress in fine-tuning video diffusion models for 3D camera control. MotionCtrl (Wang et al., 2024e) pioneered camera control by conditioning pre-trained video models with camera poses. Follow-up works (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) represent cameras as Plücker coordinates for pixel-wise conditioning. A parallel line of work (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) investigates training-free camera control of video diffusion models. More recently, SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) demonstrated impressive camera-controlled video synthesis using

large-scale synthetic training data. While these works achieve compelling results, their output is not grounded in 3D. Our work is complementary to this broad line of research: we tackle feed-forward 3D generation from camera-controlled video models, enabling novel-viewpoint rendering *after* generation. Concretely, we build upon GEN3C (Ren et al., 2025), a recent camera-controlled video diffusion model, and use it as a teacher within a self-distillation framework for 3D reconstruction; in principle, any sufficiently 3D-consistent camera-controlled video model could play this role.

Feed-forward 3D models. Early work on feed-forward 3D reconstruction mainly targeted object-centric scenes without background, directly predicting a NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) from either text or an image. More recent works (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) broaden the setting to scene-level 3D reconstruction, typically regressing per-pixel 3D Gaussians from one or more input images. However, most scene-level 3D reconstruction methods remain tied to their training distribution, e.g., RealEstate10K (Zhou et al., 2018), and generalize poorly to generated scenes. Another line of methods extends feed-forward 3D reconstruction models toward generating scenes. Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder to generate pointmaps alongside the multi-view images, which are then used for feed-forward 3D reconstruction. Closely related to our work, Wonderland (Liang et al., 2025a) uses a camera-controlled video model to generate 3D Gaussians with a feed-forward network. In contrast to these works, we use a camera-controlled video model as a teacher within a self-distillation framework to train our student 3D model *without requiring real-world multi-view data*, which lets the approach scale to diverse scenes. Importantly, we show that the same framework extends with only minor changes to feed-forward 4D scene generation, a task that has remained largely unexplored to date.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

Our key idea is to distill the implicit 3D knowledge embedded in video diffusion models into an explicit 3DGS decoder that can produce high-quality 3D representations across a wide variety of scenes. To this end, we build a general teacher–student framework in which the video diffusion model (teacher) generates RGB videos that supervise the 3DGS decoder (student) operating in the same latent space, as illustrated in Fig. 2. In this section, we detail our video diffusion model backbone (Sec. 3.1) and its role in self-distillation for the 3DGS decoder (Sec. 3.2). We then discuss the design choices for the 3DGS decoder in Sec. 4 and outline the minimal changes needed to adapt the same pipeline to dynamic 3D scenes in Sec. 5.

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Since we supervise our 3DGS decoder solely with the camera-controlled video diffusion model, we rely heavily on its 3D consistency. For this reason we instantiate our approach with GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model, though the framework is agnostic to the specific choice of a sufficiently 3D-consistent backbone. In the following we provide an overview of the key design choices made in GEN3C.

Spatiotemporal 3D cache. To improve video consistency and camera control precision, GEN3C constructs a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from input image(s) or videos, where each $\mathbf{P}^{t,v}$ is a colored point cloud obtained by unprojecting the depth estimation (Wang et al., 2024a) of an RGB image captured from camera viewpoint v at time t . The cache is organized as an $L \times V$ array, where L denotes the number of frames (temporal length) and V denotes the number of camera views.

Rendering and structured guidance. To leverage this cache, GEN3C renders each point cloud from arbitrary given camera poses, producing RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ via $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where $\mathcal{R}$ denotes the rendering function that projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D camera plane according to the camera pose $\mathbf{C}^t$ at time t . Given a sequence of camera poses $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering all cache elements produces V videos of length L , which can be stacked into image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$ for each view v . The disocclusion masks indicate areas that the video diffusion model should fill in. These renderings serve as structured visual guidance for subsequent video generation.

Video variational autoencoder (VAE). Across diffusion-based video generation models, a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) is commonly employed to compress videos into a lower-dimensional latent space for efficient training and inference.

Given a RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, a pre-trained VAE encoder $\mathcal{E}$ encodes the video into a latent space, i.e. $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$. Both training and inference of the diffusion model are then carried out in this latent space. The final video $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is decoded with a pre-trained VAE decoder $\mathcal{D}_{rgb}$. In this paper, we adopt the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model. Specifically, the latent channel dimension is $C = 16$, the temporal dimension is $L' = (L - 1)/\tau + 1$, and the spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, where the temporal compression factor is $\tau = 8$ and the spatial compression factor is $\sigma = 8$.

3.2 SELF-DISTILLATION

We train the 3DGS decoder $\mathcal{D}_s$ as a student under a teacher–student paradigm, where a camera-controlled video diffusion model $\mathcal{V}$ serves as the teacher. To obtain broad and diverse supervision, we curate a large-scale set of text prompts with large language models (OpenAI, 2025; Bai et al., 2023), generate images I with an image diffusion model (Black Forest Labs, 2024), and expand them into multi-view sequences using GEN3C (Ren et al., 2025). Because the supervision is generated rather than captured, it can in principle span an open-ended range of scene types.

Teacher–student setup. Given an input image I and a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ generates a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. The latent $\mathbf{z}$ is decoded along two branches: the pre-trained RGB decoder $\mathcal{D}_{rgb}$ produces video frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, while the 3DGS decoder $\mathcal{D}_s$ outputs explicit 3D Gaussians $\mathbf{G}$. Rendered views from 3DGS (*student*) are defined as $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$ and are supervised to match RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}$ (*teacher*), forming the self-distillation loop.

Multi-trajectory supervision. To broaden the viewpoint coverage available for each input image, we sample $V = 6$ camera trajectories per input image, as shown in Fig. 3. For each trajectory v , we construct a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ along with $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Passing these caches through the video diffusion model $\mathcal{V}$ yields latents $\mathbf{z}^v$, which the teacher decodes into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ learns to fuse these multiple $\mathbf{z}^v$ into coherent Gaussians $\mathbf{G}$ while filling disoccluded regions. Through this self-distillation design, the 3DGS decoder $\mathcal{D}_s$ is trained entirely from synthetic supervision provided by $\mathcal{V}$, without any reliance on a fixed capture setup. Operating in latent space enables efficient aggregation of many trajectories, while the explicit Gaussian output $\mathbf{G}$ enforces geometric consistency and supports downstream simulation and real-time rendering across a wide range of scenarios.

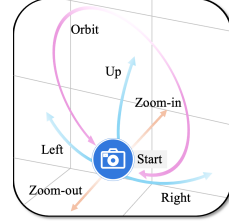


Figure 3: Six sampled camera trajectories that together broaden view coverage of the scene.

4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

Our feed-forward 3DGS decoder $\mathcal{D}_s$ transforms the synthesized multi-view latents generated by our video model $\mathcal{V}$ into an explicit 3D representation that can be rendered from arbitrary viewpoints. We adopt 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) for its explicit representation and fast rendering speed, properties that make the resulting scenes readily usable across many downstream applications. We visualize the detailed 3DGS decoder in Fig. 4 (right).

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. Previous feed-forward reconstruction frameworks generate 3D Gaussians for each pixel and are therefore limited by the number and resolution of input views they can handle. For example, GS-LRM (Zhang et al., 2024e) operates on 2–4 images at 512×512 resolution, while AnySplat (Jiang et al., 2025a) is trained with 24 images at 448×448 resolution. Our video model synthesizes $V \times L = 6 \times 121 = 726$ input views at 704×1280 resolution—far beyond the capacity of these methods, exceeding the GPU memory limit. The bottleneck lies in the attention mechanism applied to visual tokens, whose memory and compute grow with the number of pixels. To lift this limitation and make the approach scale to many high-resolution views, we avoid operating in pixel space and instead work directly in the compressed video latent space produced by our camera-controlled video diffusion model. The multi-view video latents are denoted as $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

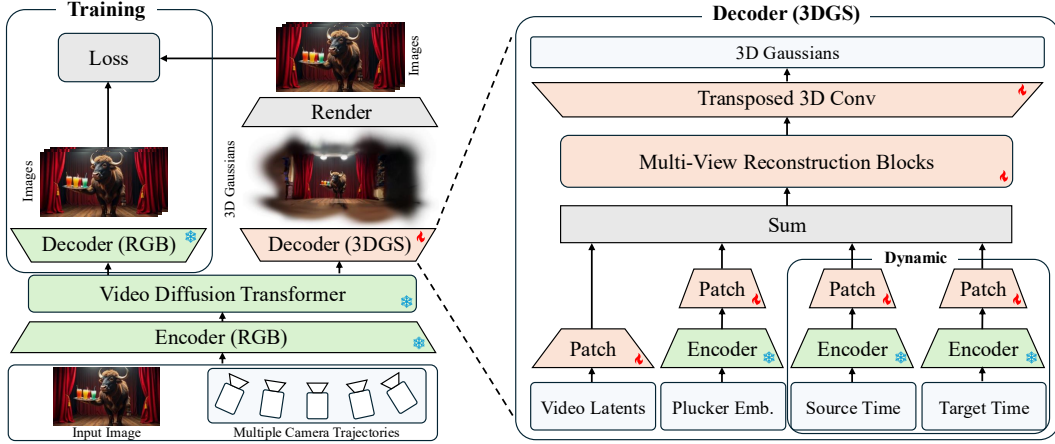


Figure 4: **3D Generative Reconstruction Framework.** Our pipeline builds upon a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large-scale data. We train a 3D Gaussian Splatting (3DGS) decoder by aligning the 2D image renderings of generated 3DGS scenes with the RGB-decoded generations of the pre-trained video model, while keeping the pre-trained autoencoder and diffusion model frozen and training only the 3DGS decoder. At inference time we use the 3DGS decoder directly, no longer requiring the RGB decoder. Time conditioning within the 3DGS decoder lets the same approach extend seamlessly from static to dynamic 3D scene generation.

Architecture. The 3DGS decoder $\mathcal{D}_s$ first maps its inputs into the hidden dimension of the main reconstruction blocks and outputs per-pixel 3D Gaussian features $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. Specifically, the inputs are the multi-view video latents $\mathbf{Z}$ and encoded Plücker embeddings $\mathbf{E}$, both with latent dimensions $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each input component is first patchified to match the hidden dimension of the main reconstruction blocks. We do not require any additional visual encoder—only a spatial 2×2 patchification layer (Dosovitskiy et al., 2021) to transform the video latents into flattened tokens for the reconstruction network. The sum of both inputs is processed through the reconstruction blocks. We follow the design introduced in Long-LRM (Ziwen et al., 2024), i.e., one block consists of one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers. We repeat the block twice to get 16 layers with 512 hidden dimensions. Finally, a transposed 3D convolution maps the hidden representation to 14 Gaussian channels: 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . Formally, the static decoder is expressed as $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. Raw Plücker embeddings are first computed from camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$. We reuse the RGB encoder $\mathcal{E}$ from the pre-trained VAE to encode Plücker embeddings. Specifically, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins using $\mathcal{E}$, and concatenate latent along the channel dimension to obtain $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, reducing the channel dimension by a factor of 2 to match video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. We supervise our 3DGS decoder with an image-based reconstruction loss. The reconstruction loss is split into a Mean Squared Error (MSE) loss $\mathcal{L}_{mse}$ and an LPIPS loss $\mathcal{L}_{lpips}$ (Zhang et al., 2018) using VGG (Simonyan & Zisserman, 2014) as a feature extractor.

Depth supervision. We observed that a decoder trained with the RGB loss alone tends to produce flattened geometry across scenes. To address this, we additionally supervise the rendered depth maps using the consistent video depth estimated by an off-the-shelf system, ViPE (Huang et al., 2025), together with the scale-invariant depth loss $\mathcal{L}_{depth}$ from Long-LRM (Ziwen et al., 2024).

Opacity-based pruning. Similar to Long-LRM (Ziwen et al., 2024), we use an L1 regularization on the opacity $\mathcal{L}_{opacity}$ and remove the Gaussians with the lowest 80% opacity.

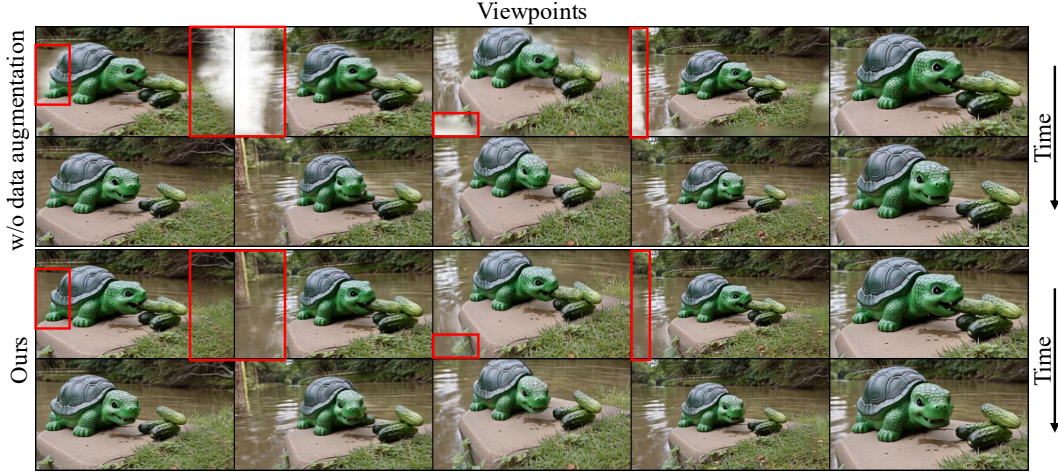


Figure 5: **Dynamic data augmentation.** Naively supervising the time-aware 3DGS decoder introduces artifacts in the generated 3D Gaussians: early timesteps seen from extreme poses exhibit low opacity in regions the supervision signal does not cover. To address this, we augment the supervision data with a motion-reversed video, so that each timestep is observed with full spatial coverage, which removes the low-opacity artifacts in early timesteps across dynamic scenes.

Total loss. Our total loss is computed as the weighted sum of all losses with weight factors λ_i

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lpips} \mathcal{L}_{lpips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where we set $\lambda_{mse} = 1.0$, $\lambda_{lpips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$.

5 EXTENSION TO DYNAMIC 3D SCENES

Our approach extends to dynamic 3D scenes with only minimal changes, demonstrating that the same framework is not restricted to static content; we outline the required modifications below.

Self-distillation for dynamic scenes. The dynamic 3D setup closely mirrors its static counterpart. Instead of a single image, the video model takes a single video of length L with corresponding camera poses as input and generates multi-view video latents that capture the same underlying motion. For video inputs, we follow a protocol analogous to the static case: we sample diverse text prompts with large language models and then generate videos using video diffusion models (Cosmos, 2025; Wan, 2025). The generated videos are subsequently annotated with camera poses and depth maps using ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. We follow the bullet-time design of Liang et al. (2025b) and design our dynamic 3DGS decoder $\mathcal{D}_s$ to generate time-dependent 3D Gaussians for all the video frames. Specifically, the dynamic 3DGS decoder $\mathcal{D}_s$ closely follows the architecture of $\mathcal{D}_s$, except that we augment the $\mathcal{D}_s$ input with the encoded source and target time embeddings $\mathbf{T}^{src}, \mathbf{T}^{tgt} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The unencoded (raw) source times are assigned to each input frame, forming $\mathbf{T}^{src, raw} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, and the target time is $\mathbf{T}^{tgt, raw} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both are spatially repeated and augmented with a 2-dimensional sinusoidal embedding to expand the channel dimension to 3, then encoded with the RGB encoder $\mathcal{E}$ to latent dimensions $\mathbf{T}^{src}, \mathbf{T}^{tgt} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The target time is repeated along the latent temporal dimension to match L' . During training, a target time step is randomly sampled, and the corresponding 3D Gaussians are decoded and supervised across all trajectories generated with dynamic data augmentation. We fine-tune $\mathcal{D}_d$ from a pre-trained $\mathcal{D}_s$ with the patchification layers of $\mathbf{T}^{src}$ and $\mathbf{T}^{tgt}$ initialized to zeros. $\mathbf{T}^{src}$ and $\mathbf{T}^{tgt}$ are directly added to the sum of $\mathbf{Z}$ and $\mathbf{E}$. Formally, the dynamic decoder is defined as $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{src}, \mathbf{T}^{tgt})$.

Dynamic data augmentation. In static 3D scenes, where time is not a factor, frames from all timesteps can jointly supervise the 3DGS renderings. For dynamic 3D scenes, however, the 3DGS changes over time, so only frames from the corresponding timestamp are valid for supervision. This restriction can lead to a trivial solution, in which the 3DGS at a given timestep simply ignores information from frames at other timesteps (see Fig. 5).

Table 1: **State-of-the-art comparisons.** We compare our method against prior works on single-image-to-3D generation across three diverse benchmarks—RealEstate10K, DL3DV, and Tanks-and-Temples—covering a broad range of real-world content.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336

To mitigate this issue, we introduce a dynamic data augmentation strategy that balances supervision across timesteps. Early frames are naturally associated with viewpoints close to the input image, whereas later frames correspond to more distant viewpoints. To counter this imbalance, we augment the training data with paired supervision views for each timestep — one near and one far. Concretely, we reverse the input video in frame order and feed it into the video model $\mathcal{V}$, which produces six additional multi-view sequences. After reversing these sequences back to the original motion order, we obtain six trajectories that progress inward (from far viewpoints toward the input). Combined with the original six outward trajectories, this yields 12 supervision views per timestep. Importantly, this augmentation is applied only during training: the reversed trajectories act purely as extra supervision signals to prevent collapse from pruning artifacts and are not required at inference time.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. To train our model, we deliberately avoid any existing multi-view datasets and instead construct our own—which we call the *Lyra* dataset—relying solely on our video model to supervise the 3DGS decoder. The 3D reconstruction setup uses 59,031 images, while the 4D setup uses 7,378 videos. All data originate from diverse text prompts and span a broad range of scenarios, including indoor and outdoor environments, humans, animals, and both realistic and imaginative content; because the data are prompt-driven, this coverage can be extended to essentially arbitrary scene categories. We synthesize six camera trajectories per image (3D) or video (4D), yielding 354,186 videos for 3D and 44,268 videos for 4D.

Baselines. We compare our method against state-of-the-art approaches for generative single-image-to-3D reconstruction, namely ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Note that *no source code is available* to run these methods on our out-of-distribution evaluation set, so we primarily rely on the quantitative comparisons reported in the respective papers.

Evaluation. Following prior work, we evaluate our model on single-image-to-3D generation using three widely used benchmarks that together cover a broad range of real-world content: RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks and Temples (Knapitsch et al., 2017). We adopt the evaluation protocol outlined in Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b), and we report performance using the standard reconstruction metrics PSNR, SSIM, and LPIPS.

6.2 MAIN RESULTS

Quantitative results. We report quantitative evaluations for single-image-to-3D in Tab. 1. Our method outperforms previous works on every benchmark and across all metrics, indicating that the gains hold consistently rather than being specific to any single dataset. This suggests that further quality improvements will come primarily from stronger video generative models, since our reconstructions inherit their advances directly. We provide additional quantitative comparisons on the *Lyra* dataset in Appendix C.

Qualitative results. We visualize novel-view renderings from our generated 3DGS scenes in Fig. 1 and Fig. 6; video results will be released upon acceptance. Across both 3D and 4D settings, our



Figure 6: **Image-to-3DGS Generation.** We visualize five views from generated 3DGS scenes spanning a diverse set of content.



Figure 7: **Ablations.** We visualize ablation results by rendering the same extreme novel viewpoint after image-to-3DGS generation, a challenging setting that stresses generalization; depth visualizations and further ablations are provided in Appendix C.4.

method generates high-quality novel-view content in unseen regions while remaining consistent with the input image or video. Additional qualitative comparisons on the *Lyra* dataset are shown in Appendix C.

6.3 ABLATIONS

In Tab. 2, we motivate our design choices by ablating key components on a diverse set of out-of-distribution prompts, chosen to probe how well each choice generalizes beyond the training distribution. We compare 3DGS renderings against videos generated by our camera-controlled video diffusion model.

Real data only. Instead of self-distillation, we train a model using only real multi-view datasets, i.e., RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b). Although these datasets are widely used in prior work, relying on them alone yields limited generalization to the broad range of out-of-distribution scenes we target.

Self-distillation + real data. Perhaps surprisingly, jointly training with self-distillation and real data does not improve over self-distillation alone. This confirms that self-distillation already provides supervision diverse and consistent enough to learn a reconstruction model that generalizes broadly, without leaning on captured datasets.

No depth loss. The depth loss prevents flat geometry and even slightly improves image-based metrics. We visualize depth maps in Appendix C.4.

No opacity pruning. We keep all Gaussians rather than pruning them by opacity. Learning

Table 2: **Ablation study on the diverse *Lyra* dataset.**

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224
Data			
real data only	19.08	0.659	0.413
self-distill. + real data	24.74	0.823	0.236
Loss			
w/o depth loss	24.31	0.811	0.247
w/o opacity pruning	24.55	0.820	0.237
w/o LPIPS loss	23.74	0.766	0.370
Architecture			
w/o multi-view fusion	17.73	0.632	0.446
w/o Mamba-2	24.58	0.818	0.241
w/o latent 3DGS		OOM	

to prune most of the 3D Gaussians makes the output representation more compact and slightly improves visual quality. Rendering at $H = 704$ and $W = 1280$ takes 18ms with pruning versus 30ms without, a $1.67\times$ speed-up.

No LPIPS loss. Adding the LPIPS loss improves robustness to input inconsistencies and better preserves high-frequency details, which helps across the varied content we consider.

No multi-view fusion. A naive alternative generates 3D Gaussians for each camera trajectory independently and then fuses them into a single point cloud. Instead, we observe clear benefits from learning multi-view fusion across trajectories within our reconstruction blocks, where each token attends to the others.

No Mamba-2. Replacing our joint Transformer/Mamba-2 design with Transformer-only blocks yields slightly lower quality. A single forward pass with $V = 6$, $L = 121$, $H = 704$, and $W = 1280$ takes 3213ms with the joint blocks versus 20922ms with Transformer-only blocks, a $6.5\times$ speed-up.

No latent-based 3DGS. Previous approaches that operate in pixel space, such as BTimer (Liang et al., 2025b), are restricted to 12 input frames, whereas our latent-space design accommodates up to 726 input frames. Consequently, running the 3DGS decoder in pixel space rather than latent space runs out of memory, underscoring why the latent formulation is what allows the method to scale to many views.

7 CONCLUSION

In this work, we propose *Lyra*, a general 3D and 4D generation framework that relies only on a single image or video input. Rather than collecting multi-view datasets, we introduce camera-controlled video diffusion models as teachers within a self-distillation framework for a student 3DGS decoder. The decoder operates in the latent space of the video model and directly reconstructs 3D Gaussians without any post-processing or optimization. This design broadens generalizability and coverage across a wide diversity of scenes, and our dynamic extension shows that the same framework transfers to 4D generation from monocular video—suggesting applicability well beyond the specific settings we evaluate. At present, the scale and consistency of our generated scenes are bounded by the capacity of the camera-controlled video diffusion model that serves as the teacher; as such models improve, our framework will directly inherit their gains and enable larger-scale, more consistent scene synthesis. Adapting auto-regressive techniques (Chen et al., 2024a) into our framework for large-scale generation is a promising direction for future work. Finally, modelling motion and tracking information within the reconstruction network, as in concurrent work (Lin et al., 2025), could further improve visual motion quality across dynamic scenes.

8 ETHICS STATEMENT

Like all generative AI technologies, our method carries risks of misuse, including the production of misleading or synthetic 3D/4D content. While such risks exist, the primary goal of this research is to contribute to simulation, robotics, and embodied AI—and, more broadly, to any field that benefits from controllable synthetic 3D content—by providing scalable and controllable tools for data generation. In this spirit, we expect our most significant contributions to arise from the following capabilities and insights.

- **The ability to enable realistic simulation for embodied agents:** We provide controllable and physically consistent camera motion within generated 3D and 4D environments, supporting the training and evaluation of agents that must perceive, navigate, and interact with a wide range of complex scenes.
- **Improved scalability for data engines:** By removing the need for multi-view capture and expensive per-scene optimization, our method helps produce high-fidelity, customizable scenes at scale—an especially good fit for robotics, reinforcement learning, and closed-loop simulation, and readily transferable to other data-hungry domains.
- **Better technical understanding:** As an academic study, this work offers a foundational case study of how camera control interacts with spatiotemporal generative models, with insights that we expect to benefit computer vision, graphics, and embodied AI more generally.

While we caution against deceptive or wrongful use of this technology—where realistic generative outputs could be used to create misleading or unverifiable media—we believe that adopting safeguards such as provenance tracking, dataset documentation, and sound evaluation practices will help ensure that generative models advance science and society in positive ways across the many settings in which they are deployed.

9 REPRODUCIBILITY STATEMENT

To fully reproduce our results and accelerate future research across the broad range of 3D and 4D generation settings this work touches, we will release our training and inference code, model weights, and data for both 3D and 4D generation upon acceptance.

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A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section provides additional information on the GEN3C (Ren et al., 2025) video diffusion model backbone introduced in Sec. 3.1 and describes how we further improve its 3D consistency, a property that matters across the diverse scenes we target.

Conservative mask refinement. A key limitation of GEN3C’s forward warping is background leakage, where occluded object parts in the source view (e.g., the lion body in Fig. 8) become visible in novel viewpoints but are not properly indicated in the disocclusion mask. This results in incomplete foreground completion by the video diffusion model, as shown in Fig. 8 (second and fourth columns), an issue that can arise for any scene with substantial occlusions.

To address this, we adopt a conservative strategy: since the geometry of occluded regions in $\mathbf{P}^{t,v}$ is unknown, we assume all such areas are occupied and derive refined disocclusion masks accordingly. This lets the video model reason about potentially visible foreground regions that standard point-based rendering would miss.

Our approach constructs a triangular mesh from camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting spatially adjacent pixels, similar to Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. This surface mesh serves as a boundary between visible and invisible regions. For each rendered pixel, we compare the standard point-based depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained via ray-surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where $\mathbf{M}_{\text{orig}}^{t,v}$ denotes the original disocclusion mask from forward warping and ϵ is a small tolerance factor. When the mesh surface is closer than the original points, we conservatively mask out these regions as potential foreground disocclusions. This yields more reliable guidance for video generation, particularly under the large viewpoint changes common across diverse scenes, as shown in Fig. 8 (third and fifth columns).

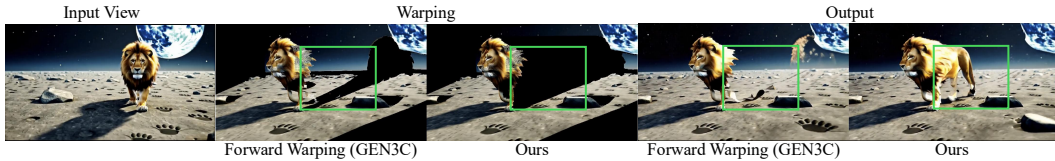


Figure 8: **Rendering function comparison.** Our model builds upon GEN3C (Ren et al., 2025), which uses forward-warped images as camera-control conditioning. We visualize forward warping (left) versus our conservative rendering function (right). The highlighted region is incorrectly filled with background pixels under forward warping but properly masked by ours, enabling correct completion—an improvement that matters wherever occlusions are present.

B MODEL DETAILS

Progressive training. Directly training the 3DGS decoder on 6 trajectories of 121 frames at 704×1280 is expensive, so we train the network progressively, as shown in Tab. 3. We further initialize the dynamic model from the final static pre-trained model, which lets a single training recipe serve both settings. Total training takes 6 days on 8 NVIDIA A100 80 GB GPUs. We use gsplat (Ye et al., 2025) as the 3DGS implementation.

Subsampling Gaussians. Generating pixel-aligned 3D Gaussians scales linearly with spatial and temporal resolution. Our generated scenes represent 726 frames at 704×1280 resolution, amounting to 654,213,120 per-pixel Gaussians. We therefore generate only one 3D Gaussian per 8×8 spatial

Table 3: **Progressive training setup.** A single progressive schedule is used across both the static and dynamic settings.

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1–6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

neighborhood, reducing the total by a factor of 64 to 10,222,080. Our opacity-based pruning then further reduces this to 2,044,416, keeping the output compact even for such large scenes.

Joint Transformer-Mamba blocks. We adopt the reconstruction block design introduced in Long-LRM (Ziwen et al., 2024): each block consists of one Transformer layer followed by seven Mamba-2 layers, and we repeat the block twice to obtain 16 layers with 512 hidden dimensions. Notably, our network is considerably smaller than those used in Long-LRM (Ziwen et al., 2024) or Wonderland (Liang et al., 2025a), which use 24 layers with 1024 hidden dimensions. While pure Transformers are typically more expressive, they are substantially slower to train and run than Mamba-2. Since we trained both configurations under the same budget, we hypothesize that Transformer-only blocks would need considerably more iterations to fully catch up in rendering quality.

Camera encoding. Following prior work (Kant et al., 2024; Zhang et al., 2024e), we use Plücker embeddings for camera encoding. The embeddings are computed at full spatial and temporal resolution and then projected into the video latent space using the pre-trained RGB encoder. Since the RGB encoder expects 3 input channels but Plücker embeddings are 6-dimensional, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins, and concatenate the resulting latent encodings along the channel dimension.

Time encoding. Analogously, to make the 1-dimensional time input compatible with the 3-channel input expected by the RGB encoder, we concatenate a 2-dimensional sinusoidal embedding to the original time value along the channel dimension. We then replicate these values across the spatial dimensions before encoding the time into the video latent space. This lets us independently encode both the source and target times into the compressed latent representation, reusing the same encoder machinery for the dynamic setting.

Dynamic data augmentation. We visualize the dynamic data augmentation procedure for two example camera trajectories in Fig. 9. The procedure creates pairs of supervision views that observe the scene from the same motion state but from opposing extreme viewpoints, broadening the spatial coverage available for each timestep.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We additionally compare our model against BTimer (Liang et al., 2025b), a recent approach for joint 3D and 4D reconstruction from posed images or videos. We reached out to the authors of BTimer, who ran experiments for us on the same scenes as *Lyra*. Since BTimer is purely regression-based, we integrate it with our GEN3C (Ren et al., 2025) video diffusion backbone: we use the same camera trajectories as in our model and provide RGB-decoded videos from GEN3C as input to BTimer. BTimer operates in pixel space and requires 12 input images. As discussed in the main paper, pixel-space models such as BTimer run out of memory when directly using the 726 high-resolution frames generated by GEN3C. To enable a fair comparison, we uniformly subsample GEN3C frames. For static evaluations, we select 12 frames that evenly cover the available viewpoints. For dynamic evaluations, we sample 11 frames to span the viewpoint range and additionally include the bullet-time frame corresponding to the motion being reconstructed, so that the target motion state is always

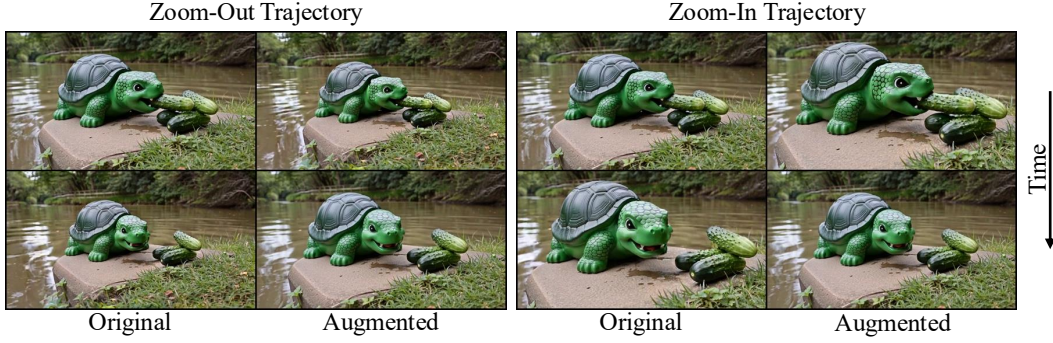


Figure 9: **Dynamic data augmentation videos.** We augment the supervision data with a motion-reversed video, so that each timestep is observed with full spatial coverage and low-opacity artifacts in early timesteps are avoided. We show two representative trajectories—zoom-out and zoom-in—together with their corresponding augmented videos, whose camera motion is flipped.



Figure 10: **Image-to-3DGS Comparison.** We compare our method with BTimer (GEN3C), visualizing five views from generated scenes. Across these varied scenes, our method exhibits markedly fewer artifacts and higher fidelity.

present as a reference. We denote this integrated baseline as BTimer (GEN3C), i.e., BTimer combined with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For image-to-3DGS generation, we present qualitative comparisons with BTimer (GEN3C) in Fig. 10 and quantitative results in Tab. 4a. Across these diverse scenes, our method consistently yields higher quality and fewer artifacts.

C.3 DYNAMIC 3D EVALUATION

BTimer (GEN3C) comparison. For dynamic 3D evaluation, we compare our method with BTimer (GEN3C), as discussed in Sec. C.1, on 100 out-of-distribution videos drawn from our dynamic *Lyra* dataset, which span a broad range of motion and scene types. We crop the video to 512×512 for a fair comparison, since BTimer was primarily trained at that resolution. As reported in Tab. 4b, our method substantially outperforms BTimer (GEN3C).

Table 4: Quantitative results across the static and dynamic *Lyra* datasets.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427	BTimer (GEN3C)	20.29	0.687	0.315
Ours	24.92	0.834	0.183	Ours	23.07	0.779	0.231

(a) Comparison on the static *Lyra* dataset.

(b) Comparison on the dynamic *Lyra* dataset.

C.4 DEPTH VISUALIZATION

We visualize depths from our generated 3DGS in Fig. 11. Depth supervision prevents flat geometries without sacrificing visual quality, and we find this benefit to hold consistently across the varied scenes we consider.

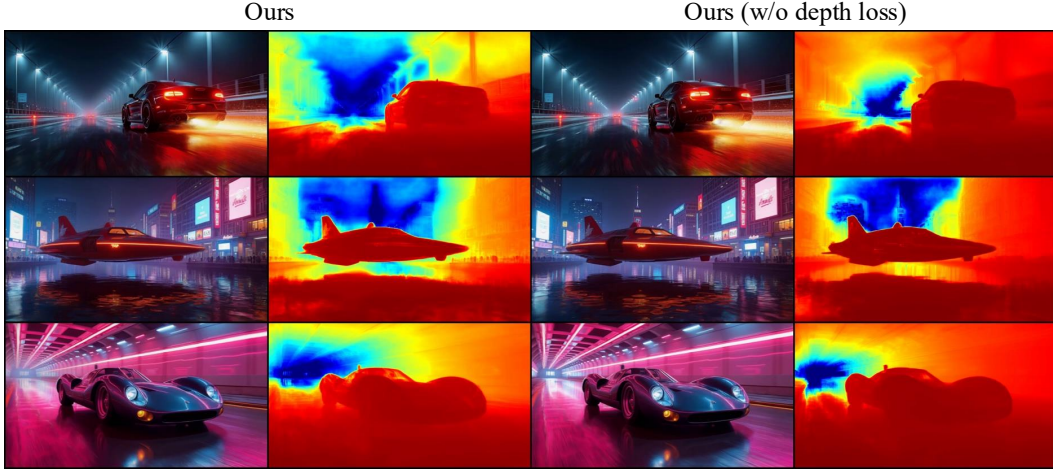


Figure 11: **Depth loss ablation.** We visualize 3DGS renderings and their corresponding depth maps. Adding the depth loss as supervision prevents flat geometries without sacrificing visual quality, a benefit that holds across the diverse scenes we generate.

D ROBOT SIMULATION

A primary motivation for building a diverse 3D generator is its ability to synthesize simulation environments for autonomous agents, and this use case illustrates how directly the generated scenes transfer to downstream applications. To demonstrate this, we develop an end-to-end pipeline: we first generate 3D scenes from text with our model, export the 3D Gaussians as a .ply file, and convert the .ply files into .usdz format. We use 3DGUT (Wu et al., 2025a) and its export function to create the .usdz files. The resulting .usdz file can then be imported into the NVIDIA Isaac robot development platform as a physically based virtual environment, within which AI-based robots can be trained and tested under a wide range of conditions. Early demo results will be released upon acceptance; we include a screenshot of the interface in Fig. 12.

E RELATED WORK

In this section, we discuss additional related work. We highlight the differences from prior works CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b) in Fig. 13.

3D generation. Early approaches to 3D generation focused primarily on single object categories, extending GANs into the 3D domain by leveraging neural renderers as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-EI et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). With the introduction of CLIP-based supervision (Radford et al., 2021), the field moved toward more flexible and diverse asset creation, enabling both text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). More

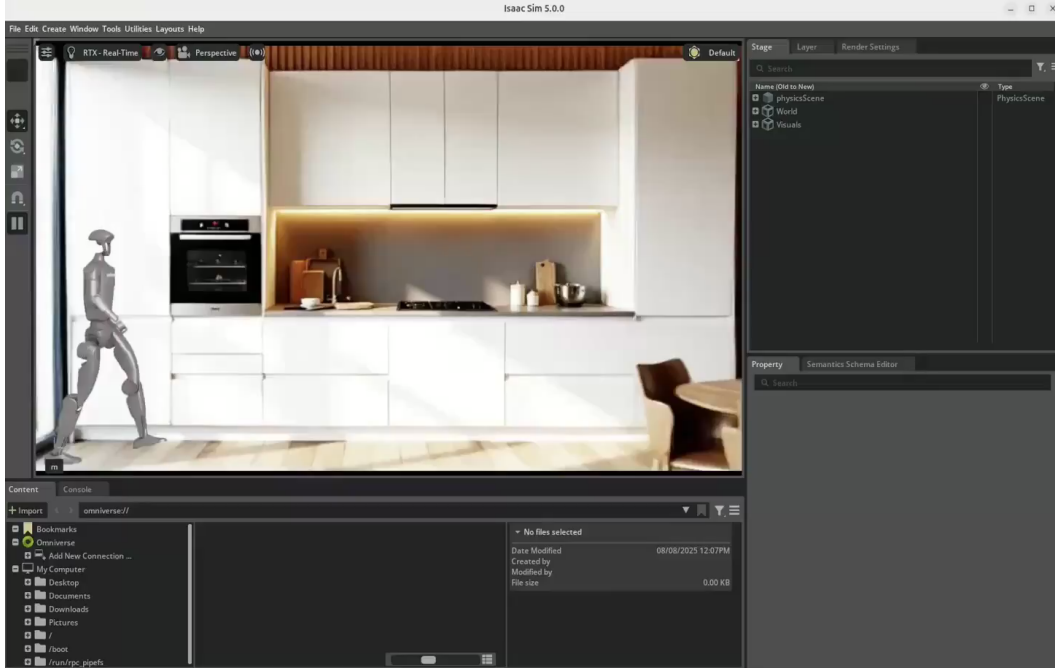


Figure 12: **Robot Simulation.** We visualize a frame of a robot simulation within the Isaac Sim 5.0 framework, using a generated 3DGS scene from our method as input—one example of the many downstream applications the generated scenes can support.

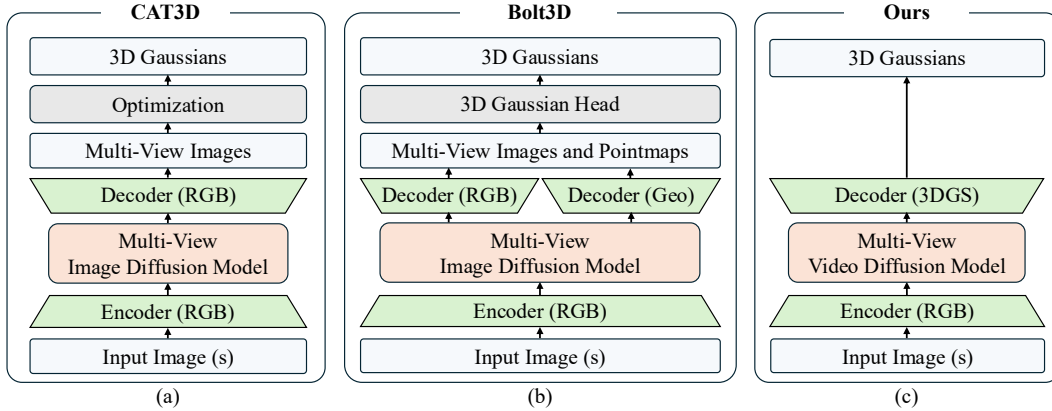


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) proposes a multi-view image diffusion model that outputs images from novel viewpoints, which are then reconstructed into 3D Gaussians via optimization. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to output pointmaps using a geometry autoencoder and, instead of relying on optimization, predicts the 3D scene with a feed-forward 3D Gaussian head. In contrast, our work builds upon a pre-trained camera-controlled video diffusion model and *directly* decodes the multi-view video latents into 3D Gaussians, a design that applies broadly across scene types.

recently, diffusion models have substantially improved quality by replacing CLIP guidance with Score Distillation Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al., 2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

To further improve structural coherence, several methods enforce multi-view consistency by generating scenes from different perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b),

while others adopt iterative inpainting for scene synthesis (Höllein et al., 2023; Shriram et al., 2024). A further line of work lifts 2D images into 3D using NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based representations coupled with diffusion models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. Complementary efforts investigate fast, feed-forward techniques that directly predict 3D content from images or text inputs (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e;d; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). In contrast to our method, however, these approaches are generally confined to producing static 3D scenes, and some target only narrow domains such as faces (Kirschstein et al., 2025). Concurrent work (Liang et al., 2025b; Xu et al., 2025) handles real scenes but does not extend to diverse generated scenes or large viewpoint changes—precisely the broad regime our framework is designed to cover.

4D generation. Recent years have seen rapid progress in 4D generation, i.e., dynamic 3D scene synthesis, across an increasingly broad set of inputs. Many approaches use text prompts or images as guidance. Following the introduction of large-scale generative models for this task (Singer et al., 2023), subsequent works have made notable advances in both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

While text conditioning is widely used, a complementary family of methods lifts 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Still other efforts incorporate physics priors into the generation pipeline (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025) or enhance motion controllability with template-driven approaches (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b), and a parallel direction emphasizes compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

Another strand of research extends 3D GAN frameworks into the 4D setting by training directly on 2D video data (Bahmani et al., 2023a; Xu et al., 2023). Such methods, however, are typically constrained by small-scale, single-category datasets that limit generalization, and most remain object-centric, often neglecting background elements. As a result, their overall visual fidelity lags behind the high photorealism of recent video generation models—the very models our method leverages to attain broad scene coverage.

Camera-conditioned video models. Recent progress has focused on incorporating camera control into video diffusion models across a growing variety of backbones. The pioneering work MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent efforts (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) improved conditioning by representing cameras with Plücker coordinates, while others follow MotionCtrl in simply flattening poses (Wang et al., 2025b). A further line of research (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) enables camera motion control without introducing additional trainable parameters. Despite these advances, all of the above rely on U-Net-based architectures, and yet other approaches adopt low-rank formulations for camera control (Zhang et al., 2025b).

Several recent works continue to extend U-Net-based approaches for camera control (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), while others begin exploring diffusion transformers. Cheong et al. (Cheong et al., 2024) propose one such transformer-based framework, though its scene and visual quality remain limited. DimensionX (Sun et al., 2024d) introduces joint space-time control in diffusion transformers but relies on pre-defined,

1566 non-continuous camera trajectories. Other works (Huang et al., 2024a; Mai et al., 2025; Jiang et al.,
1567 2025b) instead explore pose and geometry estimation with a video DiT. CAT4D (Wu et al., 2025b)
1568 proposes a multi-view video diffusion model obtained by fine-tuning a multi-view backbone. Large-
1569 scale synthetic data generation further boosts performance, as demonstrated by SynCamMaster (Bai
1570 et al., 2025b) and ReCamMaster (Bai et al., 2025a), which achieve strong results on camera-controlled
1571 video generation with transformer-based architectures. Finally, 4DiM (Watson et al., 2024) trains a
1572 space-time diffusion model from scratch for novel-view synthesis from a single input image. A related
1573 line of work edits the camera movement of videos with underlying motion via camera retargeting (Seo
1574 et al., 2025; Ma et al., 2025). A popular and broadly adopted direction conditions camera-controlled
1575 video generation on a spatio-temporal cache, e.g., built from point clouds or optical flow (Wu et al.,
1576 2025c; Ren et al., 2025; Wang et al., 2025d; Cao et al., 2025; YU et al., 2025; Jin et al., 2025; Xing
1577 et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu et al., 2025).